Chicago Taxi Demand Analysis: Optimization Through Analytics

Fleet Strategy & Predictive Roadmap

Team 2 | Grp 8 | Data Analytics Internship | July 25, 2026

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The Blindspot

Core Issue:

- Fleet managers currently lack granular visibility into peak profitability windows and payment-driven income discrepancies.

Business Impact:

Inefficient staging leads to significant lost revenue during the 5 PM surge. Furthermore, under-reporting of tips in cash transactions discourages driver retention in a competitive market.

The Lens

Project Goal:

Identify lucrative day/hour combinations and distance-based segments to provide actionable staging and digital payment recommendations for 2023 fleet operations.

Success Metrics

Peak ID: Pinpoint 5 PM revenue maximums.

Tip Discrepancy: Quantify 93% delta between card/cash reporting.

Yield ROI: Benchmark revenue per trip.

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Data Architecture & System Inputs

BigQuery Public Cloud Dataset	Taxi Company Registry
Data: Chicago Taxi Trip Records	**Data:** Operational Metadata
Volume: ~5.1M Rows	**Volume:** 36 Companies
Key Features: Timestamps, Miles, Fares, Tips	**Key Features:** Fleet Size, Company Labels

Limitation Detected — Cash Transaction Blindspot

Data is limited to traditional taxis; cash tips are historically under-reported (observed 0.35% frequency) compared to credit card verification.

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The Analytical Pipeline: Raw Data to Strategy

Data Engineering



SQL-based cleaning to remove zero-mile anomalies and filter for consistent 2023 timestamps.

Feature Selection



Categorizing rides by trip duration and payment type to identify tipping correlations.

Visualization



Developing Looker Studio dashboards to visualize demand performance for fleet dispatchers.

Strategy



Translating statistical findings into staging instructions for airport and transit hubs.

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2023 Macro Telemetry: Fleet Performance Overview



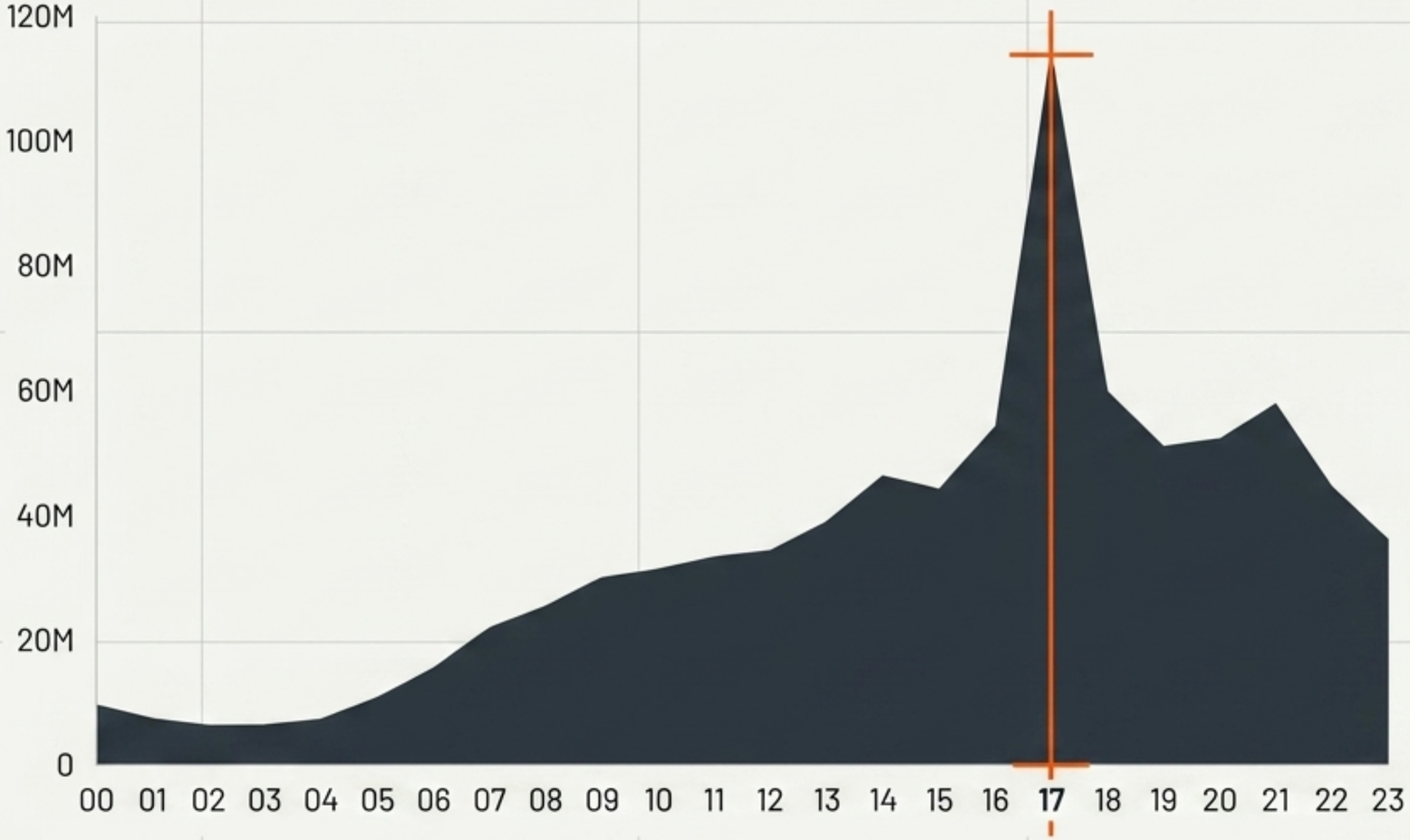
Scale: 5.8M
Total Trips

Value: $165.09M
Total Revenue

Yield: $28.50
Average Fare

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Dimension 1: The Temporal Surge



The 5 PM Surge: Demand peaks sharply at 17:00, representing the highest annual revenue hour ($12.2M).

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Dimension 2: Distance Economics



Distance Economics: Trips > 10 miles yield significantly higher average fares ($41.50) than city hops.

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Dimension 3: The Payment Discrepancy

93.63%

Credit card users tip 93.63% of the time

Digital / Card

0.35%

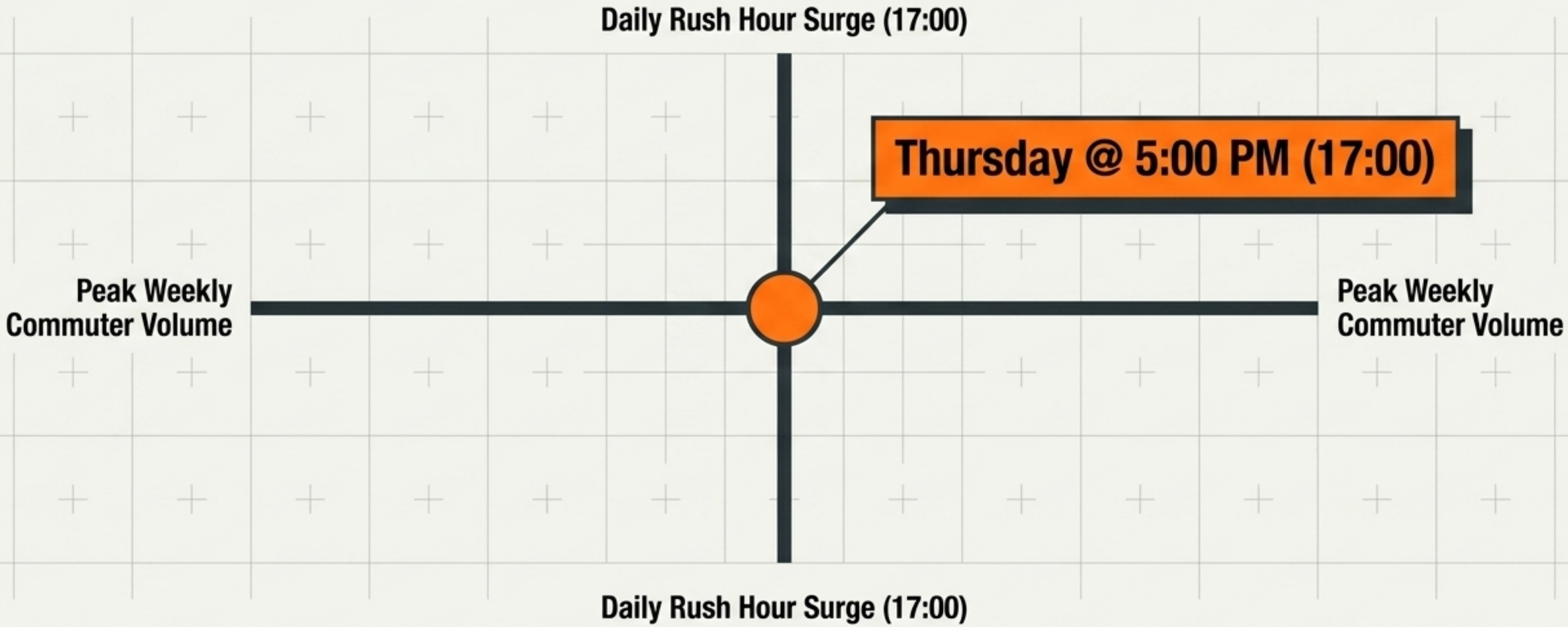
Cash tips remain statistically invisible (0.35% frequency)

Cash

The Revenue Drain: This massive reporting delta artificially depresses recorded driver income and **obscures actual fleet profitability**.

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The Crown Metric: Maximum Fleet Profitability



What is the most profitable hour-and-day combination for a Chicago taxi driver? This window captures the highest intersection of weekly commuter volume and the daily rush hour surge.

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Tactical Directives for Fleet Optimization



Optimize for 5 PM

Stagger shifts to ensure maximum vehicle density during the evening rush.



Mandate Digital POS

Functional card readers are essential to increase reported driver take-home pay by capturing the 93% tip delta.



Strategic Staging

Focus fleet presence at O'Hare and Union Station to capture high-yielding 10+ mile trips.

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Next-Gen Architecture: Machine Learning Deployment

1. Feature Engineering

Encoding categorical variables (Payment Type, Day) and scaling numerical data to prepare the raw dataset for algorithmic training.

2. Predictive Targets

Defining specific targets: Classification for predicting tip probability and Regression for forecasting exact trip fares based on distance and time.

3. Model Training

Establishing a baseline model (Logistic/Linear Regression) before deploying advanced ensemble models like Random Forest and XGBoost.

4. Full-Stack Deployment

Exporting the finaiized, high-performing model to power a live, interactive web application for real-time fare and demand predictions.

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